

Qualitative Cation Analysis

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1 Overview

In this handout, I will demonstrate the importance and usage of qualitative cation analysis. This is a qualitative skill, with little-to-no numbers attached. This relies on patterns, memory, and logic, although it is relatively simple in theory.

We begin by splitting the most important metal cations into groups. These groups aren't formed the same way the nonmetal groups are, which are by columns (noble gases, halogens, etc). These groups are formed by their general behavior towards certain reagents and conditions, as we will see later.

2 Group 1: Insoluble Chlorides

Group 1 is the smallest group, with only three cations: Ag^+ , Pb^{2+} , Hg_2^{2+} . These are also the easiest cations to determine.

These cations are very special because these are (pretty much) the only cations that precipitate with Cl^- , Br^- , I^- .

We begin by adding concentrated hydrochloric acid HCl , at concentrations above 5 M, causing AgCl , PbCl_2 , and Hg_2Cl_2 to precipitate.

2.1 Extra

There are some important things here I want to point out. First of all is the color of AgCl , which is white. You should remember that the AgBr and AgI precipitates are light-yellow and yellow, respectively. Figure 1 shows an image taken from the Wikipedia article on silver halides to demonstrate this.

Additionally, the Mohr's Method of titration is actually a very common method used to determine the amount of chloride ions in a solution. It depends on the insolubility of silver chloride AgCl , among other things. I am considering making a short handout about the Mohr's Method and its applications soon.

You probably noticed that the mercury (I) cation is written as the peculiar Hg_2^{2+} . In fact, when I first saw this, I thought that it was a typo, and that the author meant to write Hg^{2+} . As it turns out, a singular Hg^+ ion is unstable, so it will bind with another in a stable metal-metal covalent bond. For this reason, it is correct to write Hg_2Cl_2 rather than HgCl . The same is true for most other mercury (I) salts.



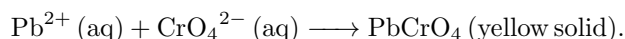
Figure 1: AgCl, AgBr, and AgI precipitates, respectively

You may have also noticed that the most important halide of all, fluoride, F^- , was left out above. This is because the group 1 cations do not precipitate with fluoride. In fact, in general, the solubility rules of fluoride are opposite of those of the rest of the halides. This can be explained using Hard-Soft Acid-Base Theory, although that is besides the point of this handout.

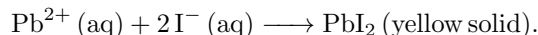
2.2 Differences between silver, lead, and mercury

Suppose you've added hydrochloric acid to your solution, yielding a white precipitate. Unfortunately, all three of these precipitates are white. There are some steps you can take to determine which of the three ions are in the solution.

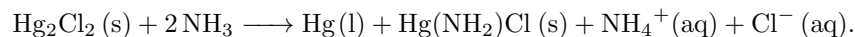
One important feature about $PbCl_2$ is that it is soluble in hot water, much more so than silver chloride and mercury (I) chloride. To determine if you have any lead (II) ions, just add some water to your precipitate and heat up the solution. After sufficient heating, isolate the solution from the precipitate. Adding a small amount of Na_2CrO_4 would cause the bright yellow precipitate $PbCrO_4$ to form.



If no such yellow precipitate is seen, this means there were no lead ions. Note that instead of sodium chromate, you can also use sodium iodide, which also yields a yellow precipitate:



After ensuring we don't have any lead ions, we have a precipitate possibly containing silver and/or mercury ions. The difference between these is their behavior when ammonia is added. Silver famously forms a stable complex: $Ag[NH_3]_2^+$. On the other hand, Hg_2Cl_2 disproportionates:



After adding the aqueous ammonia, if you see a black solid, it is a mixture of elemental mercury and the mercury (II) salt, indicating a presence of mercury in the original solution. Upon separating

the black solid from the mixture, you now just add HCl to solution. If a white solid reforms, it means you had silver as well.

Below is a summary table of Group 1.

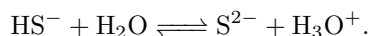
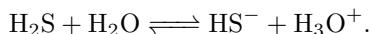
Cation	Initial precipitate	Separation step	Confirmatory test
Pb ²⁺	White PbCl ₂	Dissolves in hot water	Yellow PbI ₂ or PbCrO ₄
Ag ⁺	White AgCl	Dissolves in NH ₃	White AgCl reforms after HCl
Hg ₂ ²⁺	White Hg ₂ Cl ₂	Does not dissolve in NH ₃	Gray/black solid forms

Table 1: Summary of Group I cation tests.

3 Group 2: Acid-Insoluble Sulfides

Let's suppose we add the hydrochloric acid to our original solution and no precipitate forms. This means no group 1 cations are present. We now move on to the group 2 cations: Cu²⁺, Ni²⁺, and all the 5th or 6th period metals. As you can tell, this is a pretty large group. However, it is very very rare for you to ever need to check for the presence of something like molybdenum or hafnium, due to how rare these metals are. The most important metals in this group are Cu²⁺, Ni²⁺, Hg²⁺, Sn²⁺, As³⁺, Bi³⁺, although some of these are much more common than the rest, and there are also other metals that may show up.

To our solution with no group 1 cations, we add hydrogen sulfide gas, by bubbling it into the solution. This will cause only the most insoluble metal sulfides to precipitate. Remember that even before H₂S is added, the solution is in acidic conditions due to the added hydrochloric acid. Examine the following two equilibria:



In order for something like FeS to precipitate, we need S²⁻ ions in solution. However, because we are in acidic conditions, there is a very low amount of sulfide ions actually in solution. The large amounts of hydronium ions suppress their formation. This is why only the most insoluble metal sulfides can precipitate at this point.

3.1 Extra

Aside from essentially just copper and nickel, all the acid-insoluble sulfides are in the 5th and 6th row. This can be explained by Hard-Soft Acid-Base Theory, which I'll try to make a handout on so that I can further explain.

One slightly related thing is the existence of Sn²⁺, Bi³⁺, and others like them. Tin is in the 14th column of the periodic table, so you'd expect it to have its most common oxidation state be +4. However, due to what is called the inert pair effect, it commonly adopts the +2 oxidation state, due to its 5s electrons being inert. A similar thing is true for Bi³⁺. I'd recommend doing further research on the inert pair effect to further understand this.

3.2 Colors

Unlike the precipitates in group 1, we have more interesting colors for the precipitates of group 2. I'll touch on the most important ones.

For example, CuS is black, NiS is black or dark brown, CdS is yellow, Bi₂S₃ is black or dark brown, As₂S₃ is yellow, and Sb₂S₃ is orange. I myself haven't memorized any more group 2 sulfide colors, but you may find it interesting and/or helpful to do further research on your own.

It is important to note that these colors are not always definitive. If your precipitate is black, there's no telling if it's CuS or NiS or some other black or dark brown precipitate. Additionally, if you have multiple precipitates in solution, remember that darker precipitates will mask the color of lighter precipitates. So if your precipitate is black, it's still possible for something like As₂S₃ to be present, since its yellow color could be masked by the darker black.

There are more advanced procedures for separating the individual Group 2 cations from each other. However, this handout focuses on the major separation logic rather than every individual confirmatory test. The most important idea is that Group 2 cations form sulfides that are insoluble enough to precipitate even in acidic solution.

4 Group 3: Base-Insoluble Sulfides

After removing all group 1 and group 2 cations from our solution, we now move on to group 3. We begin by making the solution basic and then adding ammonium sulfide. We make the solution basic, usually by adding aqueous ammonia. In this environment, we are looking to form basic sulfides (or maybe some hydroxides). Here are the group 3 cations: Co²⁺, Fe²⁺, Mn²⁺, Zn²⁺, Al³⁺, Cr³⁺, and Fe³⁺.

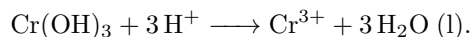
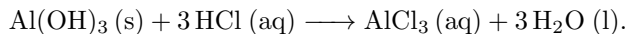
Remember how the group 2 cations included only the metals that could form sulfide precipitates in acidic conditions? We are now in basic/neutral conditions, so more precipitates can form. Specifically, we get CoS, FeS, MnS, ZnS. As you can see, the difference between the group 2 sulfides and the group 3 sulfides is a matter of solubility. For example, using the K_{sp} values, MnS needs approximately 10^5 more times S^{2-} in solution to begin precipitating than CdS does. This is why CdS can precipitate in acidic conditions and belongs in group 2, while MnS needs basic conditions and belongs in group 3.

Let's talk about the colors of these group 3 sulfides. ZnS is one of the few white sulfides. CoS and FeS are black. MnS is actually a unique light pink.

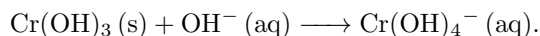
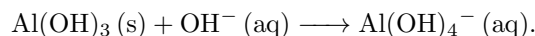
You may notice that I haven't talked about Al³⁺ or Cr³⁺ or Fe³⁺ yet. These three are different than the rest of the group 3 metals. They don't precipitate as their sulfides, instead precipitating as hydroxides: Al(OH)₃, Cr(OH)₃, and Fe(OH)₃. They also have the interesting colors of white, green, and reddish-brown, respectively.

4.1 Extra

A cool thing to remember is that $\text{Al}(\text{OH})_3$ and $\text{Cr}(\text{OH})_3$ are amphoteric, meaning that they dissolve in both acidic and basic conditions. In acidic conditions we have:



In basic conditions, we have:



$\text{Al}(\text{OH})_4^-$ is called aluminate.

Also, notice that we have a buffer solution, due to the ammonium sulfide and ammonia. This controls the pH.

5 Group 4: Insoluble Carbonates

Unlike the last two groups, which were kind of complicated, this group is pretty simple. This group consists of Ba^{2+} , Sr^{2+} , Ca^{2+} , Mg^{2+} , which is just (most) of the alkaline earth metals. Beryllium is left out because it is too uncommon.

After separating the group 1, 2, and 3 metals from the solution, we add sodium carbonate Na_2CO_3 to our solution. This causes the precipitates of BaCO_3 , SrCO_3 , CaCO_3 , MgCO_3 to form. All of these are white precipitates.

5.1 Differentiating Within Group 4

As you might guess, we use the flame test colors to tell the difference between these alkali earth metals. Barium has a green flame test, strontium has a deep crimson-red flame test, and calcium has a lighter brick-red flame test. Magnesium does not exhibit a useful or distinguishable flame test color, so some other test would need to be done to check if magnesium is present.

5.1.1 Flame Tests

I thought it might be helpful to talk about what exactly flame tests are. They are a qualitative way to determine what metal is present, due to the color produced when its ion is heated in a flame.

When a metal salt is placed in a hot flame, some electrons in the metal will absorb energy, moving to higher energy levels. When they fall back down to their ground state energy, light is emitted. Different metals have different spacings of their energies, so different colors are emitted, depending on the metal. Of course, flame tests aren't perfect, since many colors can look similar to others.

6 Group 5: Soluble Cations

After groups 1-4 have been removed, we are left with a clear solution. The only common ions in this group are every alkali metal from lithium to cesium, as well as ammonium. Lithium has a vibrant pink-crimson-red flame, sodium has a yellow flame, potassium and rubidium have a similar purple flame, violetish-blue flame test. Ammonium does not have a flame test, but it can be detected by adding strong base and lightly warming the solution, causing ammonia gas to be produced.

7 Conclusion

There are some details that I probably forgot, but I think this covers pretty much almost everything you'd need to know for the Chemistry Olympiad. If you have any questions or comments, or maybe you just want to tell me how much you love (or hate) this handout, feel free to let me know. You can contact me on Discord (bluewater16), Instagram (lishan.prado), or LinkedIn (Lishan Prado).